

Unit-5 : Research and Practical Applications

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Content

- Research and Practical Applications
- Review and Discussion of Key Topics (Refresher)
- Analyzing Real-World OSN
- Case Studies
- Research Paper Discussions on Online Social Networks and Security
- Future Trends in OSN Security and privacy

Research and Practical Applications:-

Online social networks (OSNs) significantly shape research and practical applications by enabling knowledge dissemination, collaboration, and real-world influence, though security and privacy challenges persist

Research Impacts-

OSNs amplify research visibility through platforms for sharing publications and findings, fostering global collaborations among scientists. They facilitate data collection via user interactions, aiding studies on information spread and behavior, while enabling participatory science projects. Privacy concerns, like data breaches, drive specialized research into secure OSN architectures.

Research Impacts Step-by-Step Evolution

Step 1: Knowledge Dissemination – Researchers post advances on networks like Twitter or LinkedIn, reaching peers and the public beyond traditional journals.

Step 2: Collaboration Building – Connections form via groups and conferences, leading to joint projects and co-authorships.

Step 3: Feedback and Iteration – Public comments refine studies, incorporating diverse insights for improved methodologies.

Step 4: Data-Driven Analysis – OSN metrics on user engagement inform models of viral spread and influence dynamics.

Methodologies assess OSN impact on academic research

Methodologies for assessing the impact of online social networks (OSNs) on academic research draw from bibliometrics, altimetric, and qualitative evaluations, often integrating OSN-specific data like shares and mentions. These approaches quantify visibility, collaboration, and influence while addressing limitations like data noise. They build on general research impact frameworks adapted for digital platforms.

Key Methodologies

- Altimetric: Track OSN engagement (e.g., Twitter mentions, Facebook shares) alongside downloads and views to gauge broader societal reach beyond citations.
- Bibliometric Analysis: Measures citation counts, h-index, and publication volume, correlating them with OSN activity to assess amplified productivity.
- Peer and Expert Review: Experts evaluate OSN-driven collaborations or dissemination quality through case studies and interviews.

Application Steps-

- 1.Data Collection:** Gather OSN metrics (likes, retweets) and academic outputs (papers, citations) from tools like Dimensions.ai or Scopus.
- 2.Indicator Analysis:** Compute correlations between OSN exposure and impact indicators, such as increased citations post-sharing.
- 3.Mixed-Methods Synthesis:** Combine quantitative data with qualitative testimonials to establish causation, like policy influence from OSN campaigns.
- 4.Certainty Assessment:** Use frameworks like GRADE to rate evidence reliability, accounting for biases in OSN data.

Research Paper Discussions (OSN & Security)

Foundational Surveys / Taxonomies

“Privacy and security in online social networks: A survey” (Kayes et al., ScienceDirect):

Provides a taxonomy of privacy and security attacks in OSNs, categorized by attack goals, data types, and attacker capabilities.

Use to build a taxonomy slide: profile attacks, relationship attacks, content-based attacks, social engineering, infrastructure attacks.

Research Paper Discussions (OSN & Security)

Foundational Surveys / Taxonomies

“A comprehensive survey on online social networks security and privacy” (Wiley):

Classifies OSN attacks and defenses, mapping them to OSN stakeholders (providers, third parties, users).

Good basis for a comparison table in notes: attack type vs. affected stakeholder vs. typical defense.

Research Paper Discussions (OSN & Security)

Foundational Surveys / Taxonomies

“Comprehensive Analysis of Privacy Attacks in Online Social Network” (IJSSE):

Reviews different privacy attacks (inference, de-anonymization, sybil, cloning) and argues for ML/DL-based detection; suggests blockchain and smart contracts for future decentralized OSNs.

Research Paper Discussions (OSN & Security)

Specialized Research & Practical Frameworks

Privacy-preserving community detection (Hybrid filtering + probabilistic techniques):

2024 study proposes a multi-step framework for privacy-preserving community detection on large OSNs: blended data filtering, density-oriented clustering, and a new privacy mechanism that outperforms classic k-anonymity on SNAP datasets (Yelp, Football, Zachary, Dolphin).

Application: content recommendation, influence analysis, without revealing sensitive attributes.

Research Paper Discussions (OSN & Security)

Specialized Research & Practical Frameworks

Advanced OSN attack detection (Email hijacking & DNS spoofing):

2025 framework (ASNADM) combines energy consumption profiling (EComp-FP) and opcode sequence mining (ASOSA-OSM) to detect automation-based attacks in OSN ecosystems, achieving ~99.9% precision and ~31% reduction in computational overhead.

Good example of hybrid ML + systems approach; you can map to IDS concepts you already teach.

Research Paper Discussions (OSN & Security)

Specialized Research & Practical Frameworks

OSN for cyber threat intelligence (CTI) / social network analysis of CTI:

Work on using social network analysis over malware information sharing platforms (e.g., MISP) to identify clusters of victims and shared infrastructure, enabling better incident response and CTI sharing.

Fits under “OSN methods applied *for* security” rather than “OSN as target”.

Research Paper Discussions (OSN & Security)

Specialized Research & Practical Frameworks

Community analysis of cybersecurity experts on OSNs:

Recent thesis uses OSN data (Twitter/LinkedIn) to study cybersecurity expert communities, tools, and influence, with custom tooling for graph analysis.

Can be discussed as “meta-OSN security”: how expert networks themselves can be mapped and potentially attacked.

Future Trends in OSN Security & Privacy:-

Threat & Technology Trends (2025–2026)

- AI-enabled social engineering: Generative AI now produces highly realistic phishing messages, chats, and deepfake audio/video, dramatically increasing click-through and success rates on OSNs.
- AI-vs-AI defense: Defenders deploy AI-driven anomaly detection and automated SOC copilots; attackers respond with adversarial examples and polymorphic content.
- Identity as the new perimeter: Password less, phishing-resistant MFA, risk-based access, and strong lifecycle governance become baseline; OSNs increasingly rely on behavioral and biometric signals.
- Zero-trust around OSN integrations: As OSNs integrate with enterprise tools, zero-trust access, device posture checks, and DSPM (Data Security Posture Management) extend to social data.

Future Trends in OSN Security & Privacy:-

Privacy & Regulatory Trends

- Convergence of AI governance and privacy: The EU AI Act and stricter data protection laws force platforms to provide audit trails, DPIAs, and human oversight for high-risk, data-driven personalization and moderation systems.
- Cookie less and contextual targeting: Movement from behavioral tracking to contextual advertising and data clean rooms, reducing cross-site tracking but not necessarily OSN internal profiling.
- Global tightening of privacy regulations: More countries introduce data localization, explicit consent requirements, and higher penalties, impacting how OSNs store and share data.

Future Trends in OSN Security & Privacy:-

Architectural / Research Directions for OSNs

- Decentralized and blockchain-based OSNs: Research proposes blockchain-backed, immutable logging and smart-contract governed access to improve trust and confidentiality.
- Federated and on-device learning: Privacy-preserving ML (federated learning, differential privacy) for recommendation, spam detection, and OSN abuse detection without centralizing all raw user data.
- Advanced PETs in OSNs: Homomorphic encryption, secure multi-party computation, and improved anonymization for graph data (beyond k-anonymity) integrated into privacy-preserving community detection.

Future Trends and Emerging Challenges

Emerging Technologies and Security Implications

Artificial Intelligence and Machine Learning:

Opportunity: AI-powered threat detection and anomaly identification

Risk: AI-generated deepfakes and synthetic media manipulation

Risk: Algorithmic bias in automated decision-making

Opportunity: Privacy-preserving federated learning architectures

Risk: AI models trained on sensitive user data

Blockchain and Decentralized Networks:

Opportunity: Decentralized identity management reducing single points of failure

Opportunity: Immutable audit trails for compliance

Risk: Irreversible data storage reducing "right to be forgotten"

Risk: Weak governance in decentralized systems

Future Trends and Emerging Challenges

Opportunity: Smart contracts enabling privacy-preserving interactions

Internet of Things (IoT) Integration:

Wearable devices collecting continuous health and location data

Connected home devices creating passive surveillance environments

Industrial IoT introducing new attack surface areas

Edge computing distributing processing and security responsibilities

Privacy-Enhancing Technologies (PETs)

Advanced Privacy Protection Methods:

1. **Homomorphic Encryption:** Computation on encrypted data without decryption
2. **Secure Enclaves:** Trusted execution environments (TEEs) protecting sensitive computation
3. **Differential Privacy:** Mathematical framework providing privacy guarantees
4. **Synthetic Data Generation:** Creating realistic data without revealing individuals
5. **Privacy-Preserving Machine Learning:** Training models without exposing training data
6. **Zero-Knowledge Proofs:** Proving knowledge without revealing underlying information
7. **Trusted Execution Environments:** Intel SGX, ARM TrustZone enabling secure computation

Regulatory and Governance Evolution

Expected Developments:

- Harmonization of privacy laws across jurisdictions reducing fragmentation
- Mandatory security and privacy audits for major platforms
- Enhanced enforcement budgets enabling regulatory action
- Data localization requirements affecting platform architecture
- Right to explanation for algorithmic decision-making
- Mandatory incident reporting timelines tightening response windows
- Data monetization regulation limiting commercial data exploitation

Research Challenges and Open Problems

Unresolved Issues in OSN Security and Privacy:

1. **Privacy-Utility Optimization:** Balancing stronger privacy with useful analytics[5]
2. **Graph Privacy:** Protecting sensitive information in social network structures
3. **Misinformation Detection:** Distinguishing false information at scale in real-time
4. **Deepfake Detection:** Identifying synthesized media with increasing sophistication
5. **User Behavior Prediction:** Anticipating and preventing attacks based on behavioral anomalies
6. **Insider Threat Detection:** Identifying malicious actors with legitimate access
7. **Cross-Platform Privacy:** Coordinating privacy across integrated services
8. **Contextual Privacy:** Determining appropriate information sharing in different contexts
9. **Scalable Enforcement:** Applying security controls across billions of users
10. **Zero-Day Protection:** Defending against previously unknown vulnerabilities

Practical Guidelines for OSN Users

Account Security Checklist

Immediate Actions:

- Change passwords to strong, unique credentials
- Enable multi-factor authentication on all important accounts
- Review connected applications and revoke unnecessary permissions
- Check login history and remove unrecognized sessions
- Update recovery email and phone number
- Enable login alerts and suspicious activity notifications
- Review privacy settings and limit profile visibility
- Check data download options to understand what platforms know

Privacy Protection Strategies

Data Minimization:

1. Provide minimal personal information during registration
2. Use pseudonyms or separate accounts for different purposes
3. Avoid sharing sensitive attributes (health, religion, sexuality)
4. Do not link to other accounts unless necessary
5. Disable location services unless needed
6. Restrict app permissions to essentials only

Sharing Control:

1. Review default privacy settings and adjust as needed
2. Use friend lists to categorize different relationship types
3. Share different content with different audiences
4. Avoid public sharing of sensitive information
5. Consider long-term implications of shared content
6. Understand that deleted content may be archived elsewhere

Recognizing and Avoiding Common Threats

Phishing Detection:

- Verify sender identity through independent channels
- Examine URLs carefully before clicking
- Avoid downloading attachments from suspicious sources
- Check for urgency language triggering immediate action
- Look for spelling and grammar errors in official communications
- Verify requests for authentication or financial information independently

Social Engineering Prevention:

- Be skeptical of offers that seem too good to be true
- Verify job opportunities through official channels
- Do not provide personal information to unknown contacts
- Recognize emotional manipulation tactics
- Ask for verification before sharing sensitive information
- Trust your instincts when something feels suspicious

Foundational Understanding:

Online Social Networks present a complex security and privacy landscape where:

- **Design Tension:** Platforms prioritize sharing and connectivity while struggling with security and privacy protection
- **Data Value:** The personal information OSN users generate is extraordinarily valuable, creating incentives for theft and misuse
- **Evolving Threats:** Attackers continuously develop new techniques to compromise accounts, steal data, and exploit user trust
- **Privacy Paradox:** Users often express privacy concerns but engage in behaviors that expose their information
- **Regulatory Response:** Governments worldwide are implementing stricter privacy and security requirements
- **Technological Solutions:** Privacy-preserving technologies offer promise but require careful implementation
- **User Responsibility:** Individual behavior significantly impacts security and privacy outcomes

Critical Success Factors

For Individuals:

- Treat personal information as valuable and worth protecting
- Use strong, unique authentication mechanisms
- Maintain skepticism about requests for information or action
- Stay informed about emerging threats and best practices
- Engage in security awareness and continuous learning

For Organizations:

- Implement privacy-by-design architectural principles
- Conduct regular security assessments and penetration testing
- Establish clear data governance and retention policies
- Provide comprehensive security training to employees
- Respond rapidly and transparently to security incidents
- Maintain compliance with applicable privacy regulations

For Policymakers:

- Establish baseline security and privacy requirements
- Enforce existing regulations with meaningful penalties
- Enable international coordination on privacy standards
- Support research in privacy-enhancing technologies
- Balance innovation with user protection

Future Outlook

The OSN security and privacy landscape will continue evolving as:

- Platforms increase in scale and integration
- Regulatory frameworks mature and strengthen
- Attack sophistication increases with available tools
- Privacy-preserving technologies mature and deploy
- User awareness improves through education and incidents
- Emerging technologies (AI, blockchain) introduce new considerations

Open Security Issues in Online Social Networks (OSNs)

Online Social Networks (OSNs) like Facebook, Instagram, Twitter, and LinkedIn have transformed the way people communicate and share information. However, their open and interconnected nature introduces significant security vulnerabilities. These security issues impact user privacy, data protection, identity security, and cybercrime prevention.

Major Security Issues in OSNs

1 Privacy & Data Protection Risks

- ◆ Unauthorized Data Collection: OSNs collect vast amounts of personal data, often without users' full awareness.
- ◆ Data Leaks & Breaches: Cybercriminals target social platforms to steal user data (e.g., Facebook's Cambridge Analytica scandal).
- ◆ Third-Party Access: Apps and advertisers access user information, sometimes leading to misuse.
- ◆ Metadata Exploitation: Even non-private data (like friend lists, likes) can reveal sensitive patterns.

Mitigation:

- ✓ Adjust privacy settings to limit data sharing.
- ✓ Use end-to-end encryption where possible.
- ✓ Regularly review app permissions

Identity Theft & Account Hijacking

- ◆ Phishing Attacks: Cybercriminals trick users into revealing passwords via fake login pages or messages.
- ◆ Credential Stuffing: Hackers use leaked passwords from other breaches to access OSN accounts.
- ◆ Social Engineering: Attackers manipulate users to gain access to their accounts.
- ◆ Deepfake Identity Fraud: AI-generated videos or images are used for impersonation and scams.

Mitigation:

- ✓ Enable Two-Factor Authentication (2FA).
- ✓ Use unique, strong passwords for each account.
- ✓ Be cautious of suspicious messages or friend requests

Case Study:

1. Self-Sovereign Identity (SSI) with Blockchain.

Scenario: Alice wants to prove her university degree without exposing her personal details.

- ◆ Step 1: The university issues a verifiable credential and stores it on a blockchain.
- ◆ Step 2: Alice receives this credential in her digital wallet.
- ◆ Step 3: A company requests proof of Alice's degree.
- ◆ Step 4: Alice uses Zero-Knowledge Proofs (ZKP) to verify the credential without revealing unnecessary details.
- ◆ Step 5: The company confirms her degree without accessing her full academic records.
- ✅ Outcome: Alice's privacy is protected, and her credentials remain secure. Comparison of Centralized, Hybrid, and

Decentralized (SSI) Identity Models

Case Study:

2. Google Federated Login System

Scenario: A user logs into Spotify using a Google account.

1. The user selects "Login with Google" on Spotify.
 2. Spotify redirects to Google IdP for authentication.
 3. Google verifies credentials and generates an OAuth token.
 4. Spotify receives the token and grants access without a password.
- ◆ Benefit: Seamless login experience across multiple platforms.
 - ◆ Risk: If Google credentials are hacked, all linked services are compromised

YouTube video links and animations

<https://www.youtube.com/watch?v=AQCIJAif5w8>

<https://www.youtube.com/playlist?list=PLelwGZPgtiz7KNk8Uqco3Z3pkzjE3-RkE>

https://www.youtube.com/watch?v=aiPvO781YhE&list=PL5ret0CfPJLTLRSUqZ_d9jQL-bD_Oi_YN

<https://www.youtube.com/watch?v=dphLWho8xuQ>

<https://www.youtube.com/watch?v=yJYyNAGidpU>

<https://www.youtube.com/watch?v=o9gaho46Qcl>

<https://www.youtube.com/watch?v=zhCQPsdpiH4>

<https://www.youtube.com/watch?v=QdtSKYneKL8>

<https://www.youtube.com/watch?v=HGr0Q-ktAEk>

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